

wall), which are part of a larger rectangle (think of the wall itself, or in this case, your monitor). In this case, the faux 3D illusion of floating WMs is done away with because 2D surfaces—again, your monitor—are not all that good at handling the z-axis ordering among overlapping windows. So they automatically perform the resize-move drill each time you open or close a window, using a predefined algorithm to generate a non-overlapping layout from the existing windows. You may think that the automatic handling of windows in tiling WMs takes away the fine-grained control you can have over their placement in floating WMs—but once you use tiling WMs long enough, you will find that your floating layout resembles a tiling WM-generated layout anyway.

Did I mention that all window management actions in tiling WMs can be performed using only the keyboard? In fact, most tiling WMs actively discourage the use of the mouse altogether. Tiling WMs also strive for the utmost minimalism, where nothing is preconfigured or assumed to be a sensible default for the user. Configuration usually involves editing a plain text file or, in extreme cases, the source code. While this kind of minimalist approach may not be suitable for all, its main advantage is that when you are done configuring, your WM behaves exactly the way you want it to.

A few popular tiling WMs are:

- *awesome*: Written and configured in Lua.
- *xmonad*: Written and configured in Haskell.
- *dwm*: Written and configured in C.
- *i3*: Configured via a plain text file. It follows a tabular approach to managing windows.

Installing and configuring dwm under Ubuntu

dwm is an ultra-minimal tiling WM written in C. Ubuntu 11.10 offers a binary package, but it is quite pointless, since the only way to configure *dwm* is by editing its source code and recompiling. Instead, let us grab the latest source directly from the project website.

First install the dependencies needed to compile *dwm*, along with the package *suckless-tools* which contains a bunch of utilities for use with *dwm*:

```
$ sudo apt-get install build-essential libx11-dev libxinerama-dev
sharutils suckless-tools
```

Now grab the *dwm* source code from the project website, and extract it locally, as follows:

```
$ cd /usr/local/src
$ sudo wget http://dl.suckless.org/dwm/dwm-6.0.tar.gz
$ sudo tar xzvzf dwm-6.0.tar.gz
$ chown -R `id -u`:`id -g` dwm-6.0
$ cd dwm-6.0
```

Before proceeding further with compilation, have a look at the *README* file. It tells you to modify the *config.mk* file



Figure 1: Default *dwm* desktop



Figure 2: *Dmenu*

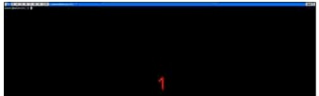


Figure 3: *XTerm* full-screen



Figure 4: Master and stack area

to match your system. However, the defaults work just fine on a normal Ubuntu system, so go ahead and compile!

```
$ sudo make clean install
```

At this point, *dwm* is installed as */usr/local/bin/dwm*, but you still need to add an entry for *dwm* so that it shows up as an option in your login screen. For this, install *dwm* from the repositories—which, as I said before, is not of much use, but it will also install the file */usr/share/xsessions/dwm.desktop*, which is what we need. So we will back up this file, remove the *dwm* package, and restore our copy of the file.

```
$ sudo apt-get install dwm
$ sudo cp /usr/share/xsessions/dwm.desktop{,.bak}
$ sudo apt-get purge dwm
$ sudo mv /usr/share/xsessions/dwm.desktop{.bak,}
```

Now we are all set to log in to our *dwm* set-up! So log out, and log back in to a *dwm* session to be greeted with a shiny new desktop that looks something like Figure 1. There you go, unabashed simplicity, right there. No oversized icons to waste screen space, no system trays, task-bars or panels to distract you and no apparent way